

Yevoo Justice Selasi – Software & AI Engineer

+233 249523573 | yevselasi@gmail.com | [GitHub](#) | [LinkedIn](#) | [X\(formerly Twitter\)](#) | Accra – Ghana

Professional Summary

Software & AI Engineer with 5+ years of experience building scalable web, mobile, and agentic AI systems across e-commerce, education, government, and enterprise domains. Experienced in architecting LangGraph/LangChain agent orchestration, including routing, memory, RAG, Deep Agents, and MCP tool integration, alongside AI-agent governance, policy engines, and production QA automation. Skilled in JavaScript, TypeScript, Python, React, FastAPI, Node.js, and PostgreSQL, with a track record of designing and delivering production-ready systems end-to-end.

Core Competencies

- LangGraph AI agent design, deterministic routing, memory/summarization, MCP tooling, pipeline reliability
- AI agent governance & QA automation, Cedar policy engines, mock-LLM testing, AI-driven Playwright generation
- Multilingual & on-device AI, Bhashini/Google Speech STT-TTS, offline Sherpa/Piper pipelines, on-device LLMs (Qwen, Gemma) across 18 languages
- Claude Skills & agentic workflow automation, custom skill design, scheduled routines, multi-step pipelines
- Full-stack development, Next.js, React, Node.js, React Native, Python, REST APIs

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL

Frameworks: React, React Native, Next.js, Node.js, FastAPI, FastMCP, Express.js, Vanilla JS, SSE

AI & Agents Engineering: LangGraph, LangChain, MCP, RAG, Guardrails (PII Redaction), Cedar Policy Engine, Claude, LangSmith, Langfuse, Bhashini STT/TTS, On-device LLM (Qwen 2.5), ML Kit (on-device translation)

Databases: PostgreSQL, MongoDB, Redis, Supabase, MinIO

Cloud & Infra: AWS (EC2, ECS), Docker, Kubernetes, Nginx, Linux, cPanel

DevOps: Git, GitHub Actions, GitLab CI/CD, rsync, Cron Jobs

APIs & Integrations: REST APIs, Google Distance Matrix API, Google Merchant Center, VideoSDK.live, Highcharts, Sanity CMS, Cloudinary

QA/Testing: Playwright (E2E, API, integration), Selenium, Allure Reporting

Professional Experience

Software & AI Engineer | Oct 2025 – Sep 2026

[BeeHyv Software Solutions](#) - Build reliable, governed AI systems on strong data foundations.

TenetXAI – AI Security & Governance | [link](#)

- Identified a critical blind spot in the policy engine: NoSQL and cloud-native databases (ClickHouse, Aerospike, Neo4j) had no detection coverage for destructive operations, unlike traditional relational databases, meaning an AI agent could delete data in these systems undetected
- Built new detection logic and Cedar policies covering destructive operations across ClickHouse, Aerospike, and Neo4j, closing the parity gap with existing relational-database protections
- Extended detection to OS-level database service termination (agents stopping database processes directly via the OS rather than through the database itself), closing 10 of 12 identified gaps with end-to-end verification and logging

AI-Assisted Portfolio Manager – Bill & Melinda Gates Foundation – Grantee Update

- Built a Claude Skills system that reads each grantee's project phase from the PO's tracker and auto-sends a tailored monthly update request with the correct template attached, eliminating manual chasing
- Built a scheduled Claude routine to detect exactly what changed in a grantee's reply, routing edits to the PO for review before the tracker updates keeping data trustworthy without manual entry
- Integrated Power Automate to handle the email round-trip end-to-end: sending requests, capturing replies, pulling attachments, and syncing everything to OneDrive automatically
- Ensured every grantee's monthly progress was captured and recorded automatically, giving POs live visibility into portfolio status without chasing updates by hand

QA Automation System

- Built a scraper that crawls a web app and generates a structured UI action contract (elements, selectors, forms), feeding an AI-driven pipeline that turns it into test plans, YAML test cases, and Playwright specs
- Triggered the full pipeline automatically on every deploy, regenerating and running the test suite so results stay accurate even when AI agents directly modify test files, removing the need for manual verification across the Agent Framework Platform
- Wired the pipeline into CI/CD as a gate, the Allure report is checked automatically after each run, and the pipeline only continues if all tests pass, catching broken components before they reach production
- Built an automated failure-triage system that classifies why a test failed (stale selector, weak assertion, bad data, real bug) and routes it to the right fix, closing the loop without manual debugging
- Built a mock-LLM service, a Dockerized fake OpenAI-compatible API replaying deterministic scripted flows, solving flaky, non-reproducible agent-workflow testing without real LLM calls

Agent Framework — AI Agent Orchestration Platform (LangGraph) | [link](#)

- Implemented cross-node shared context (context_write/context_read) across frontend and backend action nodes, enabling reliable data handoff through multi-step workflows
- Architected a Cedar policy-based governance layer for the agent platform, tenant-scoped access control (discover/invoke deny rules) across tools, models, workflows, users, and roles, with a policy builder UI so teams could target rules without writing raw Cedar
- Built agent memory management via sliding-window summarization with tool-pair repair, tuning token/threshold parameters after diagnosing context-length overflow issues in production
- Diagnosed and fixed a critical shared-context bug where LangChain MCP tool outputs weren't reaching downstream agent nodes, tracing root cause to how tool message content is structured

PUCAR & Chat SDK – LangGraph Agent Reliability & SDK Architecture | [link](#)

- Diagnosed and resolved critical disconnects between LangGraph agents and UI Action MCP server restoring production reliability
- Migrated React-based chat application into a framework-agnostic Vanilla JS SDK for seamless third-party embedding
- Rebuilt UI components using class-based Vanilla JS architecture with full state and lifecycle management
- Developed dynamic UI Actions SDK parsing backend-driven action instructions
- Implemented History API-based SPA navigation without page reloads
- Integrated Bhashini STT/TTS with Google Speech API fallback for resilient multilingual audio support

Open Agent Platform (OAP) | [link](#)

- Built end-to-end MCP server onboarding workflows (create, configure, approve, duplicate, organize, download), streamlining client setup
- Implemented multi-collection drag-and-drop organization using dnd-kit
- Developed Swagger (1.0–3.0) to OpenAPI 3.1 auto-conversion pipeline
- Optimized polling logic and background request handling improving platform stability

Bihar Krishi – Full-Stack AI Chat System (Frontend & LangGraph Agent) | [link](#)

- Migrated Bhashini STT/TTS processing to the frontend, reducing audio latency by ~50% and expanding multilingual support to 18 languages for 20M+ farmers
- Built a UI Actions system rendering backend-driven buttons/navigation from metadata, eliminating hardcoded frontend logic and enabling scalable interaction flows
- Redesigned LangGraph routing with structured normalization and fallback parsing, and disabled audio recording during LLM processing — together cutting audio/routing errors by ~90%
- Secured all query paths via token/profile-state validation and regex-based OTP safeguards, preventing false-positive session invalidations
- Packaged the chat experience as a framework-agnostic, embeddable chatbot widget, integrated across multiple third-party government platforms (Pucar, Dristi)

Lead Software Developer | Apr 2023 – Dec 2025

[Nexotin Company Ltd](#) - A fashion shopping app with media, chat, and video scrolling.

- Co-founded and led development of a React Native (Expo Router) shopping app with real-time buyer-vendor chat (Supabase), media-rich messaging (Cloudinary), and infinite video scrolling for product discovery
- Designed and maintained a scalable 100+ endpoint RESTful API (Express.js, MongoDB) powering all mobile/admin functionality, with full-text search using MongoDB text indexes
- Built a URL-shortener and SSR-powered deep-link system (Next.js) for sharing products and vendor profiles, plus a subscription-based push notification system to drive re-engagement
- Built a multi-vendor order system with split orders, real-time status updates, and delivery-fee calculation via the Google Distance Matrix API

FullStack Developer & Instructor | Feb 2022 – Nov 2025

[SFAN\(Stars From All Nations\)](#) & [ReadyforWork](#) - A career accelerator equipping African youth with in-demand digital skills.

- Built the entire platform end-to-end — student, coach, and employer portals — from architecture through deployment, while serving as Software Engineering Instructor for 2 cohorts teaching full-stack development
- Developed a dynamic student dashboard with lesson/project scores, career advisories, and progress visualization (Highcharts), plus real-time chat/video (VideoSDK.live) for coach sessions and interviews
- Created Talent Passport (Gemini AI-generated résumés from student data) and CareerPilot, which analyzes CVs and Talent Passport data to visualize student strengths as a React Flow diagram
- Built a Job Tracker System using Python/Selenium to scrape and match job listings to students based on their strengths
- Built the coach and employer platforms — expertise-based coach registration with scheduling, and employer onboarding with project posting, candidate shortlisting, and integrated interview tools

FullStack Developer | Oct 2020 – Jan 2022

[Bolstaar Company Limited](#) - Farm Equipment Rental Platform

- Built a rental platform (Next.js, React, PostgreSQL) for farmers to discover and rent equipment by proximity, with location-based search ranking
- Built a USSD server on AWS EC2 (*920150#) for order placement/callbacks, extending access to non-smartphone users

- Integrated real-time chat (Supabase) connecting farmers directly with equipment owners
- Owned backend, database, and infrastructure end-to-end — deployed via cPanel with rsync for efficient updates

Junior Backend Developer | Jan 2021 – Apr 2022

[EPDL Training](#) - Tech skills training center

- Built and maintained RESTful APIs using Node.js and Express.js for course registration, authentication, and user management, with JWT-based auth and secure session handling
- Designed and optimized PostgreSQL schemas for managing users, course data, and transactions
- Collaborated with frontend developers to ensure seamless API integration and consistent error handling

Education

B.Sc. Computer Science with Mathematics, University of Ghana (Legon)